

The Non-Analytic Critical-Region Term

Part 1: Theory

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Abstract

A pure fluid's thermodynamic state is fully determined by its reduced Helmholtz free energy $\alpha(\delta, \tau) = \alpha_0(\delta, \tau) + \alpha_r(\delta, \tau)$, where $\delta = \rho/\rho_r$ is the reduced density and $\tau = T_r/T$ the inverse reduced temperature. Near the critical point, ordinary polynomial and Gaussian-bell terms in α_r cannot reproduce the sharp, non-classical curvature of real fluid behaviour. IAPWS-95 and Span–Wagner-style equations of state add a dedicated *non-analytic* term for exactly this region. This document derives that term's structure; Part 2 covers the numerical subtleties of evaluating it, Part 3 the Rust implementation, and Part 4 walks through the verified result.

1 Why an ordinary polynomial term is not enough

Away from the critical point, α_r is well approximated by sums of

$$\alpha_r^{\text{power}}(\delta, \tau) = \sum_i n_i \delta^{d_i} \tau^{t_i} e^{-\delta^{l_i}}, \quad (1)$$

$$\alpha_r^{\text{Gaussian}}(\delta, \tau) = \sum_i n_i \delta^{d_i} \tau^{t_i} e^{-\eta_i(\delta-\varepsilon_i)^2 - \beta_i(\tau-\gamma_i)^2}. \quad (2)$$

Both forms are smooth, analytic functions of (δ, τ) everywhere. Real fluids, however, exhibit *non-classical critical scaling*: the isochoric heat capacity c_v diverges at the critical point with a power-law exponent that no finite sum of analytic terms can reproduce exactly. IAPWS-95 (and Span–Wagner-style multiparameter EOS more generally) address this with a small number of specially-shaped terms concentrated tightly around $(\delta, \tau) = (1, 1)$ — the reduced critical point — chosen so the *overall fit* closely tracks measured near-critical data without needing to solve the full non-classical scaling problem exactly [1].

2 The non-analytic term's functional form

Each non-analytic term i contributes

$$\alpha_{r,i} = n_i \Delta^{b_i} \delta \psi \quad (3)$$

built from three auxiliary functions, each itself a function of (δ, τ) :

$$\theta = (1 - \tau) + A_i [(\delta - 1)^2]^{1/(2\beta_i)}, \quad (4)$$

$$\Delta = \theta^2 + B_i [(\delta - 1)^2]^{a_i}, \quad (5)$$

$$\psi = \exp(-C_i(\delta - 1)^2 - D_i(\tau - 1)^2). \quad (6)$$

Reading these three pieces separately clarifies what each is doing:

- ψ (Eq. 6) is an ordinary two-dimensional Gaussian bell centred exactly on the critical point $(\delta, \tau) = (1, 1)$, with independent widths C_i, D_i in the density and temperature directions. It is what confines the whole term’s influence to a small neighbourhood of the critical point — away from it, $\psi \rightarrow 0$ rapidly and $\alpha_{r,i} \rightarrow 0$ regardless of the other factors.
- θ (Eq. 4) measures a kind of “distance along the critical isotherm,” combining how far τ is from 1 with a density-offset correction raised to the fractional power $1/(2\beta_i)$ — this fractional exponent is precisely the non-analytic ingredient: no integer power of $(\delta - 1)$ can produce it.
- Δ (Eq. 5) combines θ^2 with a second, purely density-dependent fractional-power term, and is itself raised to a further fractional power b_i in Eq. 3. This double layering of fractional powers is what lets the term reproduce the steep, non-classical curvature of c_v and related derivatives near the critical point that Eqs. 1–2 cannot.

For Water (IAPWS-95), CoolProp’s `Water.json` carries exactly two such terms ($i = 1, 2$), with coefficients $A_i \approx 0.32$, $B_i \approx 0.2$, $\beta_i = 0.3$, $a_i = 3.5$, $b_i \in \{0.85, 0.95\}$, $C_i \in \{28, 32\}$, $D_i \in \{700, 800\}$ [2]. The large C_i, D_i values confirm the Gaussian ψ factor is indeed very tightly peaked — its half-width in δ is only $\sim 1/\sqrt{C_i} \approx 0.19$ reduced-density units.

3 What this buys, physically

Without this term, an EOS evaluated *exactly at* the critical point $(\delta, \tau) = (1, 1)$ is missing a contribution that is small in absolute magnitude everywhere except in a narrow band around the critical point — but inside that band, it is the difference between a qualitatively wrong critical-point pressure and one that reproduces the defining IAPWS-95 value to machine precision (Part 4 shows this numerically: 5.2×10^{-14} relative error once implemented, versus a documented $\sim 10^{-4}$ residual before). The same non-analytic shape appears (with different coefficients) in Span & Wagner’s CO₂ equation of state [3] and other IAPWS/Span–Wagner-style multiparameter EOS — it is a standard building block, not a Water-specific hack.

4 What Part 2 covers

Equations 4–6 are well-defined mathematically at every $(\delta, \tau) \neq (1, 1)$, but the term needs first and second partial derivatives in δ and τ (for pressure, c_v , c_p , and the speed of sound) — several of the intermediate quantities in those derivatives are formally 0/0 or 0^0 *exactly at* the critical point. Part 2 works through the derivative chain and the numerical guard needed to evaluate it there.

References

- [1] W. Wagner and A. Pruß. “The IAPWS Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use”. In: *Journal of Physical and Chemical Reference Data* 31.2 (2002), pp. 387–535. DOI: [10.1063/1.1461829](https://doi.org/10.1063/1.1461829).
- [2] Ian H. Bell et al. *CoolProp: An Open-Source Reference-Quality Thermophysical Property Library*. <https://github.com/CoolProp/CoolProp>. Commit 0e67fe7, cloned 2026-07-05. 2026.
- [3] Roland Span and Wolfgang Wagner. “A New Equation of State for Carbon Dioxide Covering the Fluid Region from the Triple-Point Temperature to 1100 K at Pressures up to 800 MPa”. In: *Journal of Physical and Chemical Reference Data* 25.6 (1996), pp. 1509–1596. DOI: [10.1063/1.555991](https://doi.org/10.1063/1.555991).